

Large
Language
Models

Introduction to Large Language Models

Language models

- Remember the simple n-gram language model
 - Assigns probabilities to sequences of words
 - Generate text by sampling possible next words
 - Is trained on counts computed from lots of text

Large language models (LLMs) share core goals with N-gram models but differ vastly in architecture and capability. Both assign probabilities to word sequences, generate text by sampling next words, and train by predicting the next word.

Similarities

Probability Assignment: N-grams use fixed counts like $P(\text{word} | \text{prior N-1 words})$; LLMs compute dynamic probabilities via neural predictions.

Next-Word Sampling: Both generate autoregressively—N-grams pick from Markov tables, LLMs from softmax over vocab.

Training Objective: Maximum likelihood estimation; N-grams count frequencies, LLMs minimize cross-entropy loss on next-token targets.

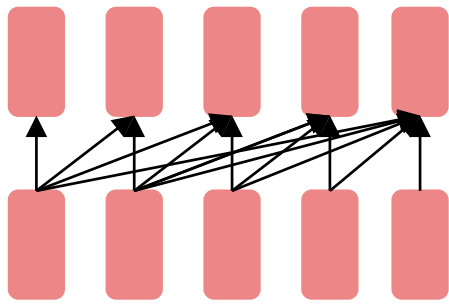
Key Differences

Aspect	N-Gram Models	Large Language Models
Context	Fixed window (N-1 prior words only)	Arbitrary length via attention (thousands of tokens)
Sparsity	Suffers unseen N-grams; needs smoothing	Dense embeddings handle novelty
Dependencies	Short-range only (Markov assumption)	Long-range via Transformers
Parameters	Table-based (vocab ^N size explodes)	Billions in weights, scales with data
Training	Frequency counts (no gradients)	Gradient descent on massive corpora

Large language models

- Even though pretrained only to predict words
- Learn a lot of useful language knowledge
- Since training on a **lot** of text

Three architectures for large language models

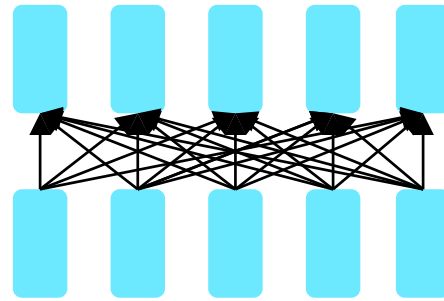


Decoders

GPT, Claude,

Llama

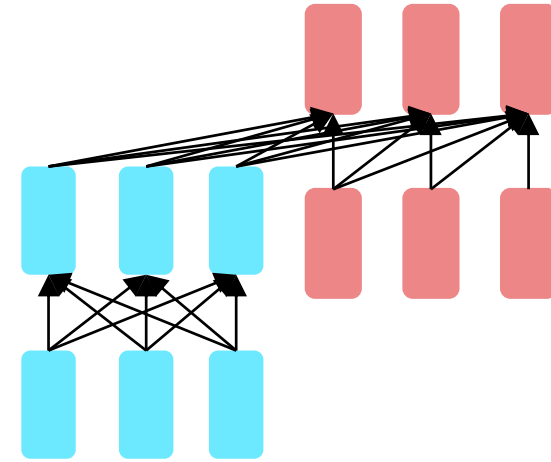
Mixtral



Encoders

BERT family,

HuBERT



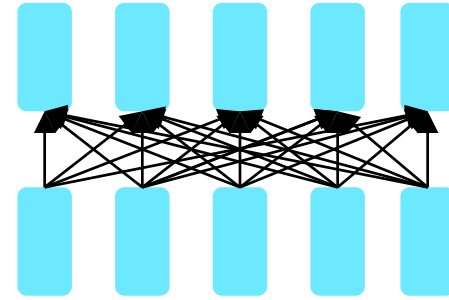
Encoder-decoders

Flan-T5, Whisper

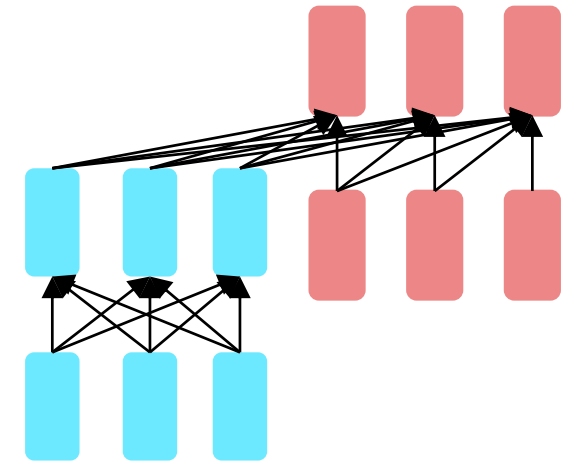
Encoders

Many varieties!

- Popular: Masked Language Models (MLMs)
- BERT family
- Trained by predicting words from surrounding words on both sides
- Are usually **finetuned** (trained on supervised data) for classification tasks.



Encoder-Decoders



- Trained to map from one sequence to another
- Very popular for:
 - machine translation (map from one language to another)
 - speech recognition (map from acoustics to words)

Large
Language
Models

Introduction to Large Language Models

Large Language Models

Large Language Models: What
tasks can they do?

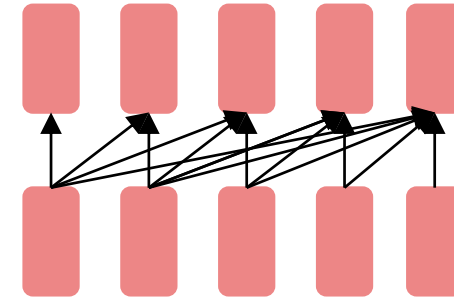
Big idea

Many tasks can be turned into tasks of predicting words!

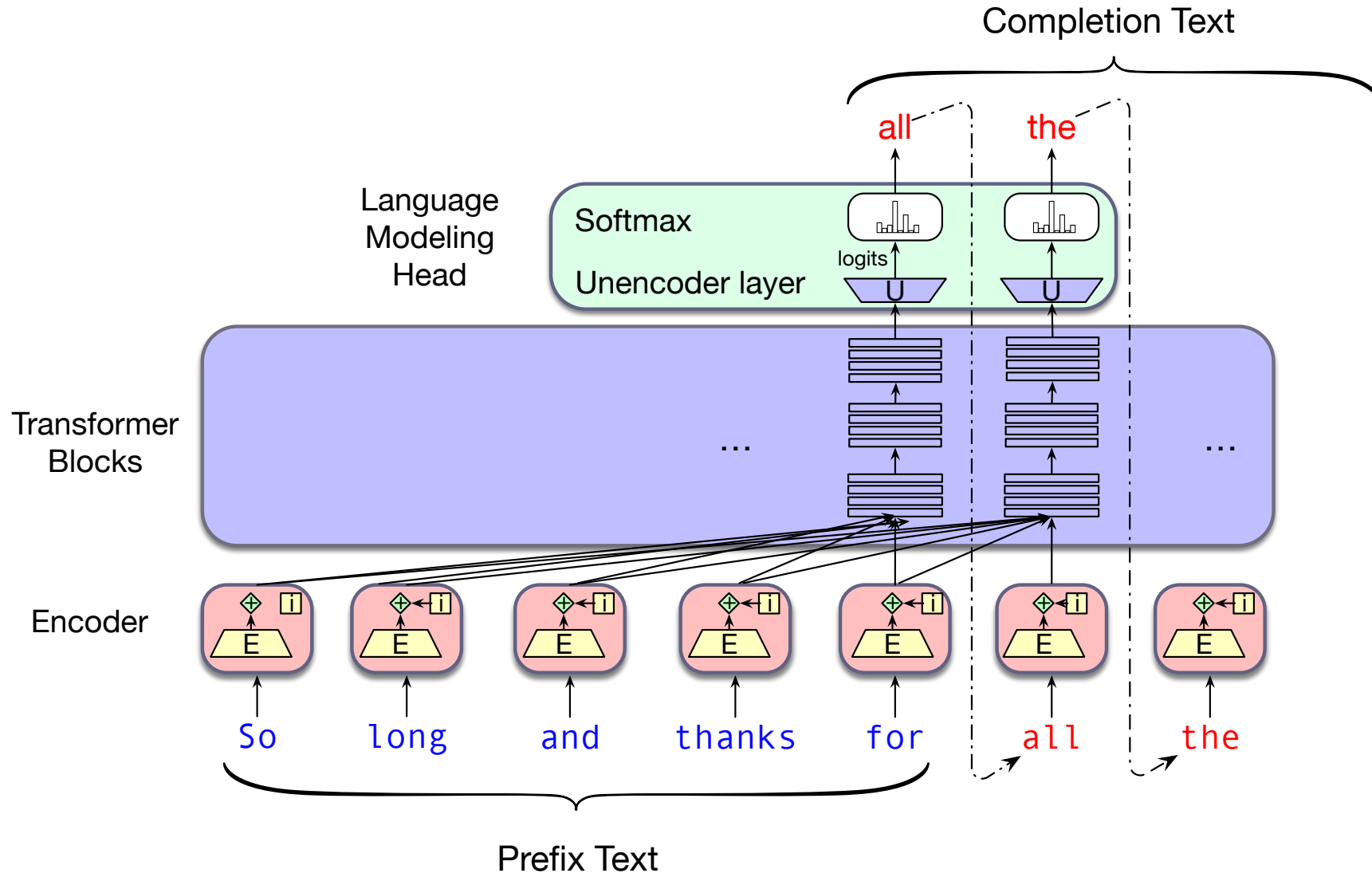
This lecture: decoder-only models

Also called:

- Causal LLMs
- Autoregressive LLMs
- Left-to-right LLMs
- Predict words left to right



Conditional Generation: Generating text conditioned on previous text!



Many practical NLP tasks can be cast as word prediction!

Sentiment analysis: “I like Jackie Chan”

1. We give the language model this string:
The sentiment of the sentence "I
like Jackie Chan" is:
2. And see what word it thinks comes next:

$P(\text{positive} | \text{The sentiment of the sentence "I like Jackie Chan" is:})$

$P(\text{negative} | \text{The sentiment of the sentence "I like Jackie Chan" is:})$

Framing lots of tasks as conditional generation

QA: “Who wrote The Origin of Species”

1. We give the language model this string:

Q: Who wrote the book ‘ ‘The Origin of Species”? A:

2. And see what word it thinks comes next:

$P(w|Q: \text{Who wrote the book ‘ ‘The Origin of Species”? A:})$

3. And iterate:

$P(w|Q: \text{Who wrote the book ‘ ‘The Origin of Species”? A: Charles})$

Summarization

The only thing crazier than a guy in snowbound Massachusetts boxing up the powdery white stuff and offering it for sale online? People are actually buying it. For \$89, self-styled entrepreneur Kyle Waring will ship you 6 pounds of Boston-area snow in an insulated Styrofoam box – enough for 10 to 15 snowballs, he says.

Original

But not if you live in New England or surrounding states. “We will not ship snow to any states in the northeast!” says Waring’s website, ShipSnowYo.com. “We’re in the business of expunging snow!”

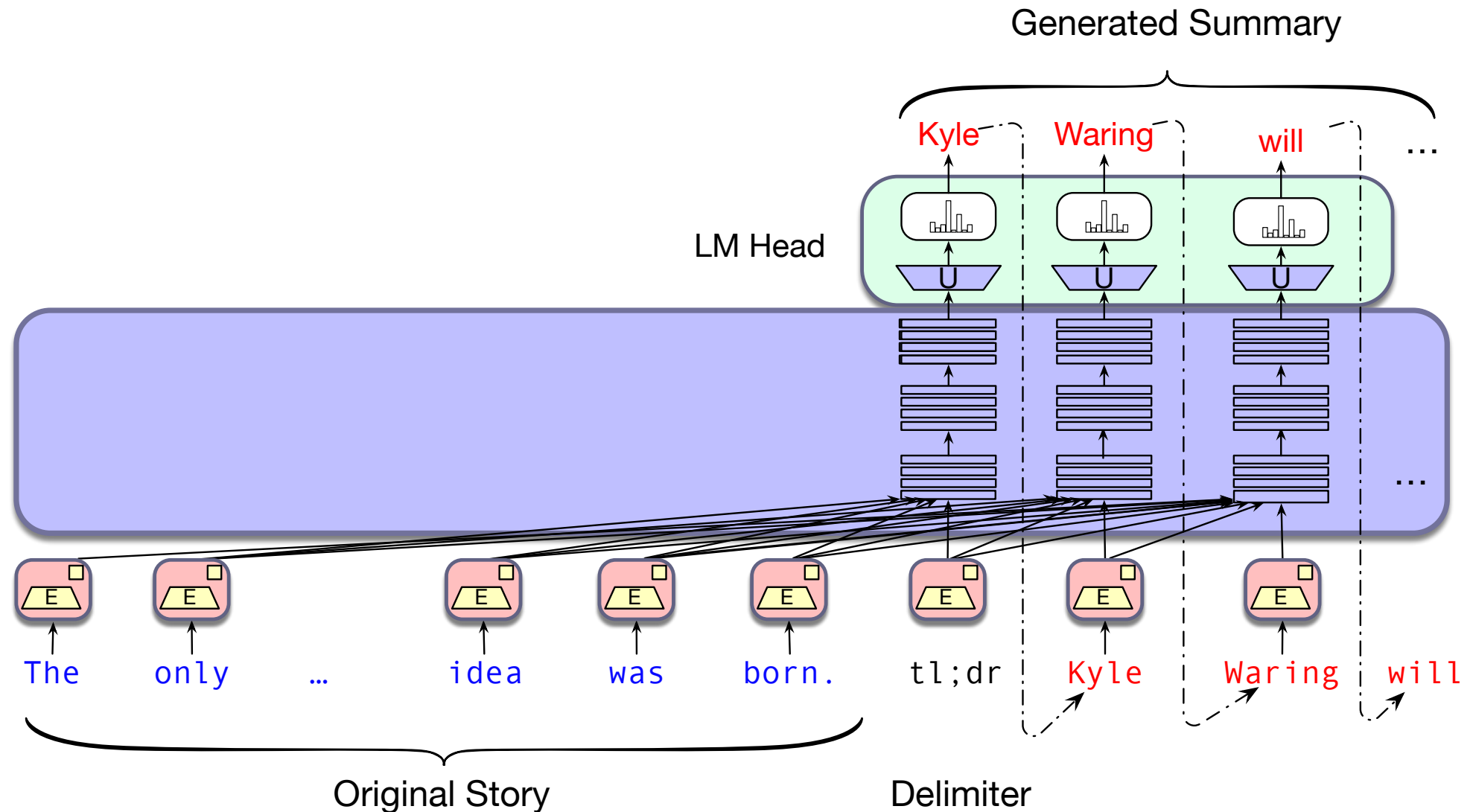
His website and social media accounts claim to have filled more than 133 orders for snow – more than 30 on Tuesday alone, his busiest day yet. With more than 45 total inches, Boston has set a record this winter for the snowiest month in its history. Most residents see the huge piles of snow choking their yards and sidewalks as a nuisance, but Waring saw an opportunity.

According to Boston.com, it all started a few weeks ago, when Waring and his wife were shoveling deep snow from their yard in Manchester-by-the-Sea, a coastal suburb north of Boston. He joked about shipping the stuff to friends and family in warmer states, and an idea was born. [...]

Summary

Kyle Waring will ship you 6 pounds of Boston-area snow in an insulated Styrofoam box – enough for 10 to 15 snowballs, he says. But not if you live in New England or surrounding states.

LLMs for summarization (using tl;dr)



Large Language Models

Large Language Models: What
tasks can they do?

Large
Language
Models

Sampling for LLM Generation

Decoding and Sampling

This task of choosing a word to generate based on the model's probabilities is called **decoding**.

The most common method for decoding in LLMs: **sampling**.

Sampling from a model's distribution over words:

- choose random words according to their probability assigned by the model.

After each token we'll sample words to generate according to their probability *conditioned on our previous choices*,

- A transformer language model will give the probability

Random sampling

```
i ← 1  
wi ∼ p(w)  
while wi ≠ EOS  
    i ← i + 1  
    wi ∼ p(wi | w<i)
```

Random sampling doesn't work very well

Even though random sampling mostly generate sensible, high-probable words,

There are many odd, low- probability words in the tail of the distribution

Each one is low- probability but added up they constitute a large portion of the distribution

So they get picked enough to generate weird sentences

Factors in word sampling: **quality** and **diversity**

Emphasize **high-probability** words

+ **quality**: more accurate, coherent, and factual,

- **diversity**: boring, repetitive.

Emphasize **middle-probability** words

+ **diversity**: more creative, diverse,

- **quality**: less factual, incoherent

Top-k sampling:

1. Choose # of words k
2. For each word in the vocabulary V , use the language model to compute the likelihood of this word given the context $p(w_t | w_{<t})$
3. Sort the words by likelihood, keep only the top k most probable words.
4. Renormalize the scores of the k words to be a legitimate probability distribution.
5. Randomly sample a word from within these remaining k most-probable words according to its probability.

Top-p sampling (= nucleus sampling)

Holtzman et al., 2020

Problem with top- k : k is fixed so may cover very different amounts of probability mass in different situations

Idea: Instead, keep the top p percent of the probability mass

Given a distribution $P(w_t | \mathbf{w}_{<t})$, the top- p vocabulary $V(p)$ is the smallest set of words such that

$$\sum_{w \in V(p)} P(w | \mathbf{w}_{<t}) \geq p$$

Temperature sampling

Reshape the distribution instead of truncating it

Intuition from thermodynamics,

- a system at high temperature is flexible and can explore many possible states,
- a system at lower temperature is likely to explore a subset of lower energy (better) states.

In **low-temperature sampling**, ($\tau \leq 1$) we smoothly

- increase the probability of the most probable words
- decrease the probability of the rare words.

Temperature sampling

Divide the logit by a temperature parameter τ before passing it through the softmax.

Instead of

$$\text{ ~~$\mathbf{y} = \text{softmax}(u)$~~ }$$

We do

$$\mathbf{y} = \text{softmax}(u/\tau)$$

Temperature sampling

$$0 \leq \tau \leq 1$$

$$\mathbf{y} = \text{softmax}(\mathbf{u} / \tau)$$

Why does this work?

- When τ is close to 1 the distribution doesn't change much.
- The lower τ is, the larger the scores being passed to the softmax
- Softmax pushes high values toward 1 and low values toward 0.
- Large inputs pushes high-probability words higher and low probability word lower, making the distribution more greedy.
- As τ approaches 0, the probability of most likely word approaches 1

Large
Language
Models

Sampling for LLM Generation

Large
Language
Models

Pretraining Large Language Models: Algorithm

Pretraining

The big idea that underlies all the amazing performance of language models

First **pretrain** a transformer model on enormous amounts of text

Then **apply** it to new tasks.

Self-supervised training algorithm

We just train them to predict the next word!

1. Take a corpus of text
2. At each time step t
 - i. ask the model to predict the next word
 - ii. train the model using gradient descent to minimize the error in this prediction

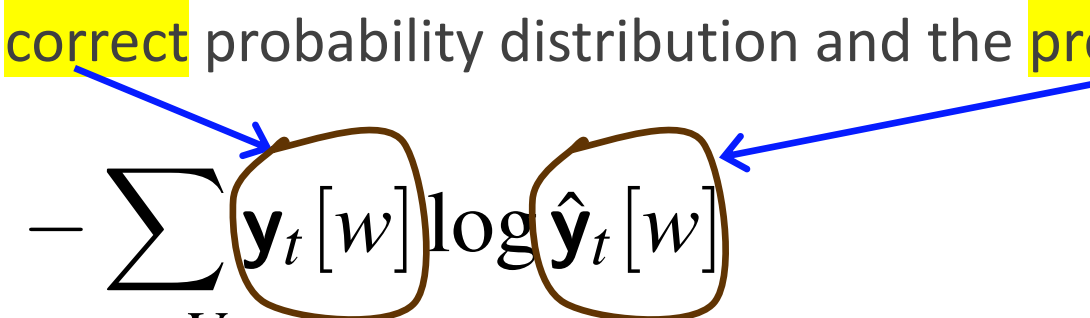
"**Self-supervised**" because it just uses the next word as the label!

Intuition of language model training: loss

- Same loss function: **cross-entropy loss**
 - We want the model to assign a high probability to true word w
 - want loss to be high if the model assigns too low a probability to w
- CE Loss: The negative log probability that the model assigns to the true next word w
 - If the model assigns too low a probability to w
 - We move the model weights in the direction that assigns a higher probability to w

Cross-entropy loss for language modeling

CE loss: difference between the **correct** probability distribution and the **predicted** distribution

$$L_{CE} = - \sum_{w \in V} \mathbf{y}_t[w] \log \hat{\mathbf{y}}_t[w]$$


The correct distribution $\mathbf{y}_t$ knows the next word, so is 1 for the actual next word and 0 for the others.

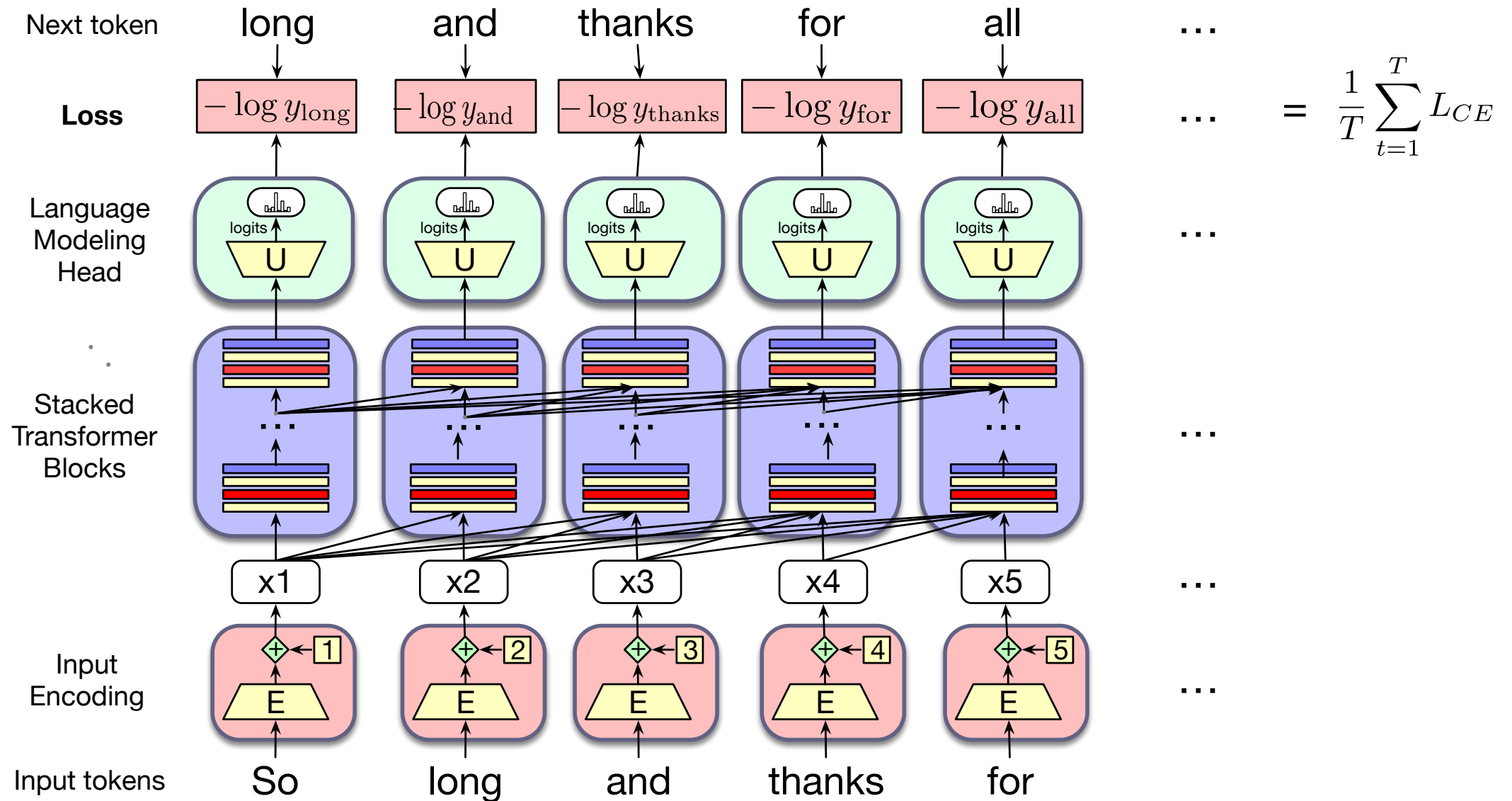
So in this sum, all terms get multiplied by zero except one: the logp the model assigns to the correct next word, so:

$$L_{CE}(\hat{\mathbf{y}}_t, \mathbf{y}_t) = -\log \hat{\mathbf{y}}_t[w_{t+1}]$$

Teacher forcing

- At each token position t , model sees correct tokens $w_{1:t}$
 - Computes loss ($-\log$ probability) for the next token w_{t+1}
- At next token position $t+1$ we ignore what model predicted for w_{t+1}
 - Instead we take the **correct** word w_{t+1} , add it to context, move on

Training a transformer language model



Large
Language
Models

Pretraining Large Language Models: Algorithm

Large Language Models

Pretraining data for LLMs

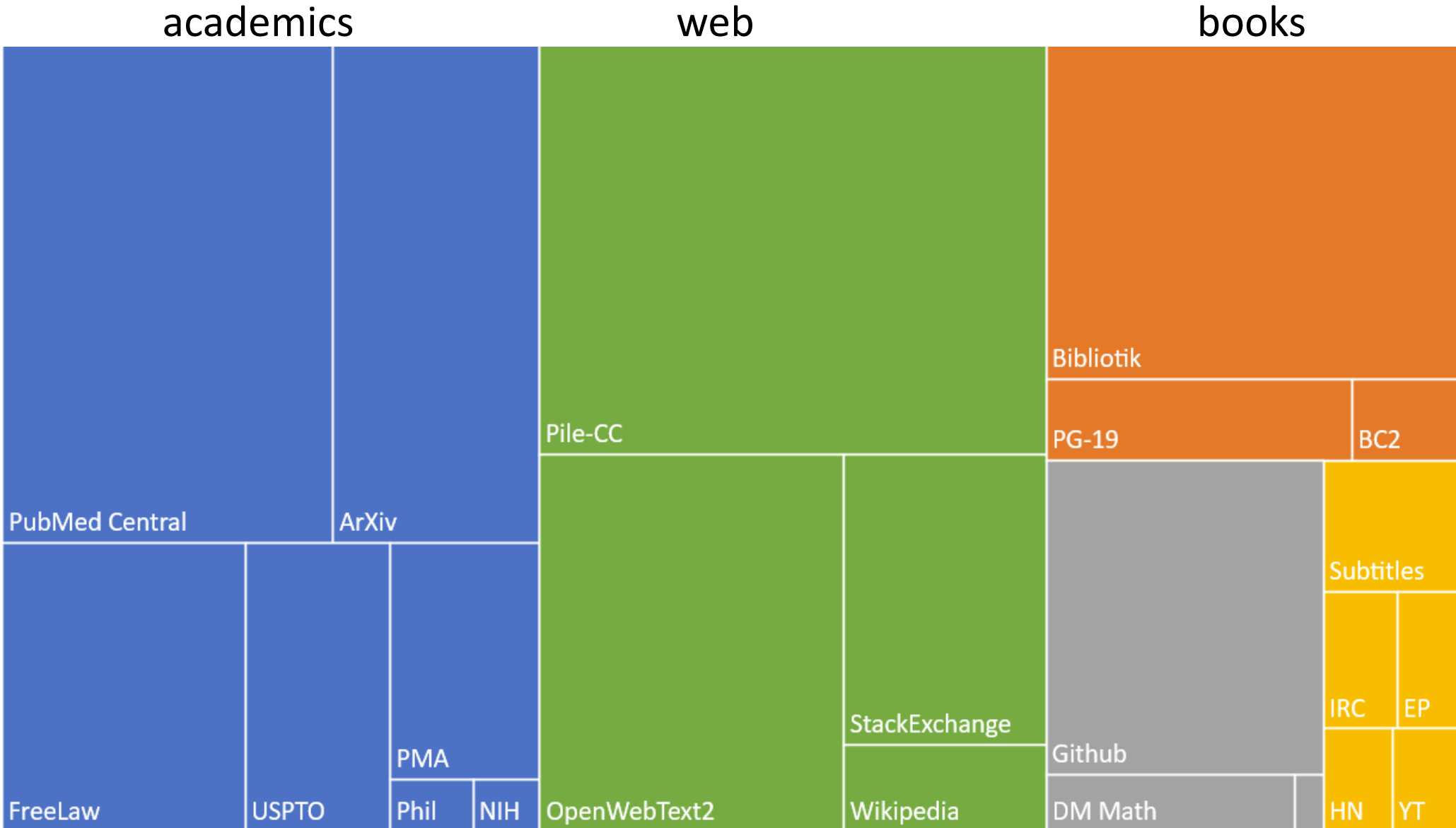
LLMs are mainly trained on the web

Common crawl, snapshots of the entire web produced by the non-profit Common Crawl with billions of pages

Colossal Clean Crawled Corpus (C4; [Raffel et al. 2020](#)), 156 billion tokens of English, filtered

What's in it? Mostly patent text documents, Wikipedia, and news sites

The Pile: a pretraining corpus



dialog

Filtering for quality and safety

Quality is subjective

- Many LLMs attempt to match Wikipedia, books, particular websites
- Need to remove repeatative, adult content
- Deduplication at many levels (URLs, documents, even lines)

Safety is also subjective

- Toxicity detection is important, although that has mixed results
- Can mistakenly flag data written in dialects like African American English

What does a model learn from pretraining?

- There are canines everywhere! One dog in the front room, and two dogs
- It wasn't just big it was enormous
- The author of "A Room of One's Own" is Virginia Woolf
- The doctor told me that he
- The square root of 4 is 2

Big idea

Text contains enormous amounts of knowledge

Pretraining on lots of text with all that knowledge is what gives language models their ability to do so much

But there are problems with scraping from the web

Copyright: much of the text in these datasets is copyrighted

- Not clear if fair use doctrine in US allows for this use
- This remains an open legal question

Data consent

- Website owners can indicate they don't want their site crawled

Privacy:

- Websites can contain private IP addresses and phone numbers

Large Language Models

Pretraining data for LLMs

Large
Language
Models

Finetuning

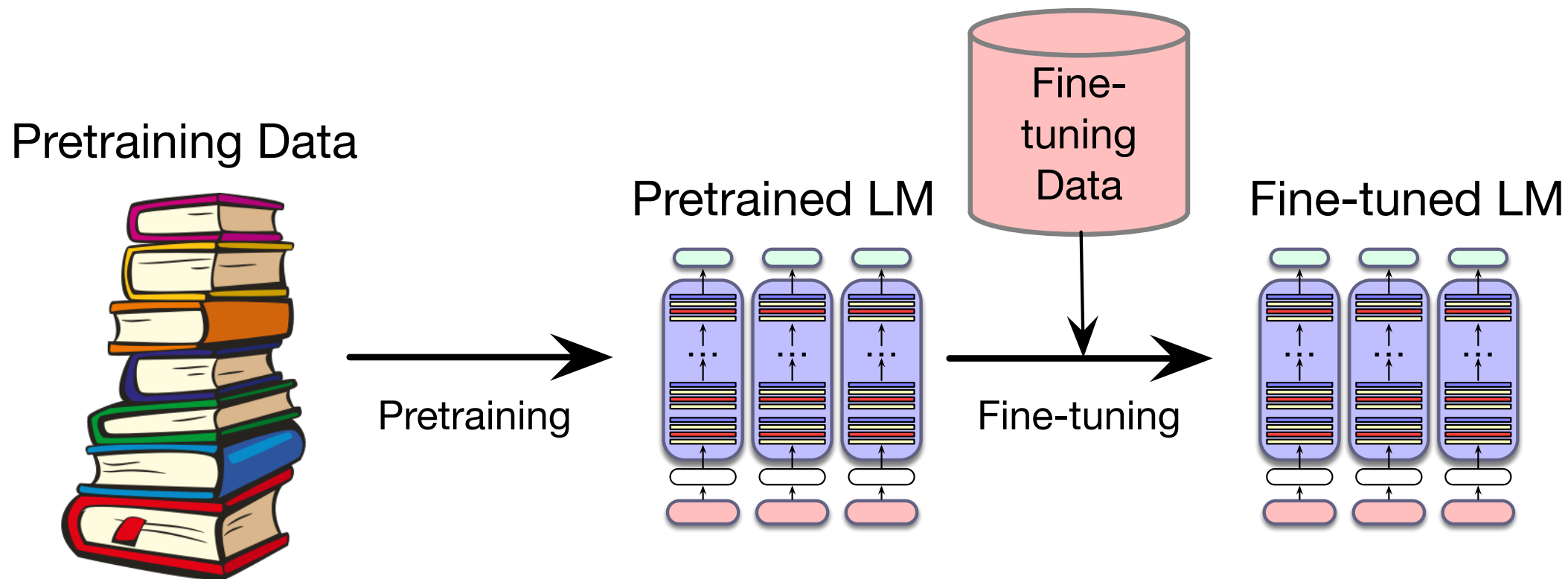
Finetuning for adaptation to new domains

What happens if we need our LLM to work well on a domain it didn't see in pretraining?

Perhaps some specific medical or legal domain?

Or maybe a multilingual LM needs to see more data on some language that was rare in pretraining?

Finetuning



"Finetuning" means 4 different things

We'll discuss 1 here, and 3 in later lectures

In all four cases, finetuning means:

**taking a pretrained model and further adapting
some or all of its parameters to some new data**

1. Finetuning as "continued pretraining" on new data

- Further train all the parameters of model on new data
 - using the same method (word prediction) and loss function (cross-entropy loss) as for pretraining.
 - as if the new data were at the tail end of the pretraining data
- Hence sometimes called **continued pretraining**

Large
Language
Models

Finetuning

Large
Language
Models

Evaluating Large Language
Models

Perplexity

Just as for n-gram grammars, we use perplexity to measure how well the LM predicts unseen text

The perplexity of a model θ on an unseen test set is the **inverse probability that θ assigns to the test set, normalized by the test set length**.

For a test set of n tokens $w_{1:n}$ the perplexity is :

$$\begin{aligned}\text{Perplexity}_{\theta}(w_{1:n}) &= P_{\theta}(w_{1:n})^{-\frac{1}{n}} \\ &= \sqrt[n]{\frac{1}{P_{\theta}(w_{1:n})}}\end{aligned}$$

Why perplexity instead of raw probability of the test set?

- Probability depends on size of test set
 - Probability gets smaller the longer the text
 - Better: a metric that is **per-word**, normalized by length
- **Perplexity** is the inverse probability of the test set, normalized by the number of words
(The inverse comes from the original definition of perplexity from cross-entropy rate in information theory)

Probability range is $[0,1]$, perplexity range is $[1,\infty]$

Perplexity

- The higher the probability of the word sequence, the lower the perplexity.
- Thus the lower the perplexity of a model on the data, the better the model.
- **Minimizing perplexity is the same as maximizing probability**

Also: perplexity is sensitive to length/tokenization so best used when comparing LMs that use the same tokenizer.

Many other factors that we evaluate, like:

Size

Big models take lots of GPUs and time to train, memory to store

Energy usage

Can measure kWh or kilograms of CO2 emitted

Fairness

Benchmarks measure gendered and racial stereotypes, or decreased performance for language from or about some groups.

Large
Language
Models

Dealing with Scale

Scaling Laws

LLM performance depends on

- Model size: the number of parameters not counting embeddings
- Dataset size: the amount of training data
- Compute: Amount of compute (in FLOPS or etc)

Can improve a model by adding parameters (more layers, wider contexts), more data, or training for more iterations

The performance of a large language model (the loss) scales as a power-law with each of these three

Scaling Laws

Loss L as a function of # parameters N , dataset size D , compute budget C (if other two are held constant)

$$L(N) = \left(\frac{N_c}{N} \right)^{\alpha_N}$$

$$L(D) = \left(\frac{D_c}{D} \right)^{\alpha_D}$$

$$L(C) = \left(\frac{C_c}{C} \right)^{\alpha_C}$$

Scaling laws can be used early in training to predict what the loss would be if we were to add more data or increase model size.

Number of non-embedding parameters N

$$\begin{aligned} N &\approx 2 d n_{\text{layer}} (2 d_{\text{attn}} + d_{\text{ff}}) \\ &\approx 12 n_{\text{layer}} d^2 \\ &\quad (\text{assuming } d_{\text{attn}} = d_{\text{ff}}/4 = d) \end{aligned}$$

Thus GPT-3, with $n = 96$ layers and dimensionality $d = 12288$, has $12 \times 96 \times 12288^2 \approx 175$ billion parameters.

A typical decoder-only LLM has these hyperparameters:

d : model (hidden) dimension

d_{attn} : total attention projection dimension (usually $d_{\text{attn}}=d$)

d_{ff} : feed-forward hidden dimension (e.g. $4d$)

n_{layer} : number of transformer blocks

Within each block you have:

1. Attention projections

Multi-head attention is implemented as four dense projections of shape $d \rightarrow d_{\text{attn}}$ or $d_{\text{attn}} \rightarrow d$:

$$W_Q \in \mathbb{R}^{d \times d_{\text{attn}}}$$

$$W_K \in \mathbb{R}^{d \times d_{\text{attn}}}$$

$$W_V \in \mathbb{R}^{d \times d_{\text{attn}}}$$

$$W_O \in \mathbb{R}^{d_{\text{attn}} \times d}$$

Total attention weights per layer:

$$\# \text{attn params} \approx 3(d \cdot d_{\text{attn}}) + d_{\text{attn}} \cdot d = 4 d d_{\text{attn}} \approx 2d(2d_{\text{attn}}).$$

Feed-forward network (FFN) projections

The FFN is usually a 2-layer MLP:

$$W_1 \in \mathbb{R}^{d \times d_{\text{ff}}}, W_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$$

Total FFN weights per layer:

$$\# \text{ff params} \approx d \cdot d_{\text{ff}} + d_{\text{ff}} \cdot d = 2 d d_{\text{ff}}.$$

Putting it together across layers

Per layer, ignoring small terms (layer norms, biases, rotary embeddings etc.), you get approximately:

#params per layer $\approx 4 d d_{\text{attn}} + 2 d d_{\text{ff}}$.

Now factor out a common $2d$:

$4 d d_{\text{attn}} + 2 d d_{\text{ff}} = 2d (2d_{\text{attn}} + d_{\text{ff}})$.

Then multiply by the number of layers n_{layer} :

$N_{\text{non-embed}} \approx n_{\text{layer}} \times 2d (2d_{\text{attn}} + d_{\text{ff}})$.

KV Cache

In training, we can compute attention very efficiently in parallel:

$$\mathbf{A} = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V}$$

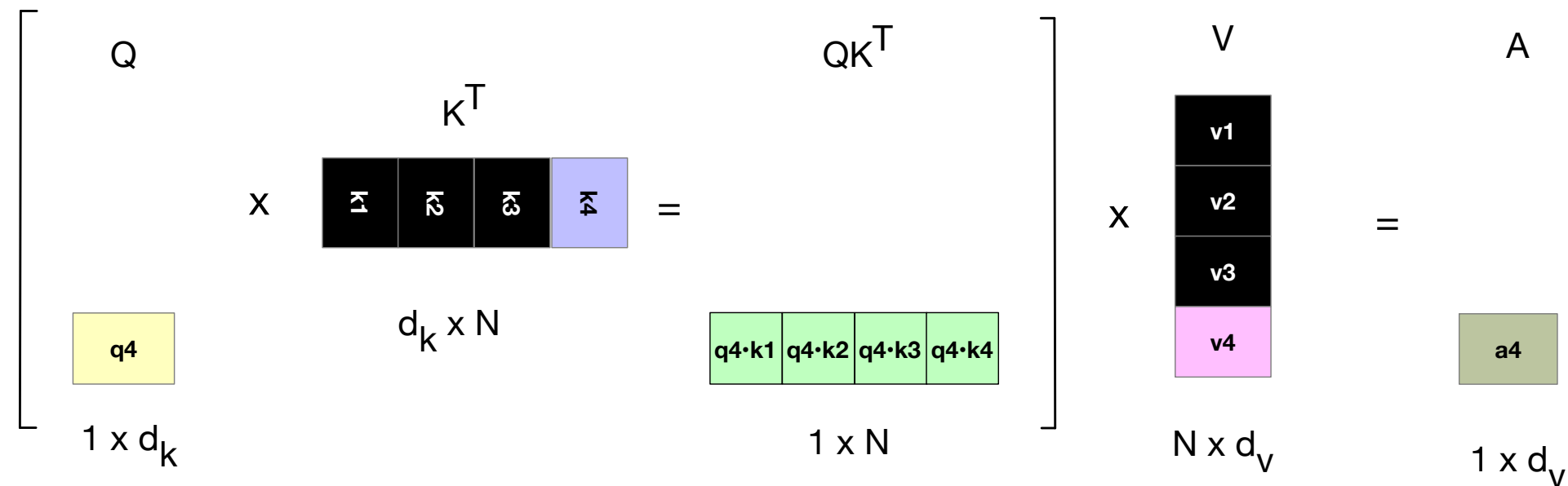
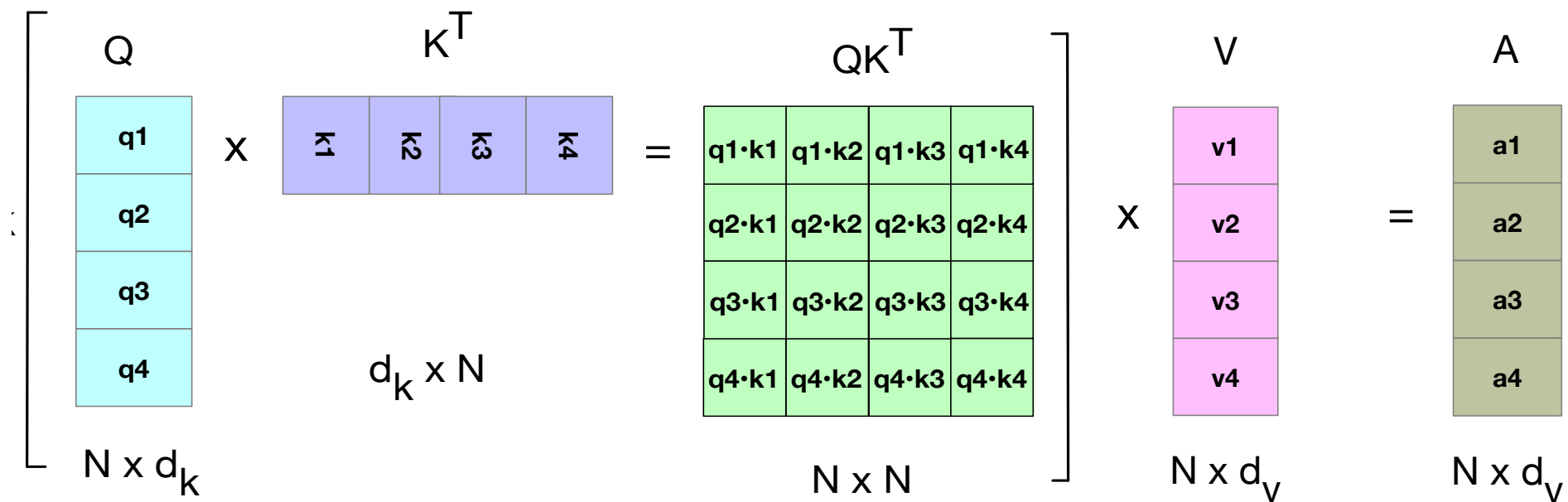
But not at inference! We generate the next tokens **one at a time!**

For a new token x , need to multiply by W^Q , W^K , and W^V to get query, key, values

But don't want to **recompute** the key and value vectors for all the prior tokens $x_{<i}$

Instead, store key and value vectors in memory in the KV cache, and then we can just grab them from the cache

KV Cache



Parameter-Efficient Finetuning

Adapting to a new domain by continued pretraining (finetuning) is a problem with huge LLMs.

- Enormous numbers of parameters to train
- Each pass of batch gradient descent has to backpropagate through many many huge layers.
- Expensive in processing power, in memory, and in time.

Instead, **parameter-efficient fine tuning** (PEFT)

- Efficiently select a subset of parameters to update when finetuning.
- E.g., freeze some of the parameters (don't change them),
- And only update some a few parameters.

LoRA (Low-Rank Adaptation)

- Transformers have many dense matrix multiply layers
 - Like W^Q, W^K, W^V, W^O layers in attention
- Instead of updating these layers during finetuning,
 - Freeze these layers
 - Update a low-rank approximation with fewer parameters.

LoRA

- Consider a **matrix W** (shape $[N \times d]$) that needs to be updated during finetuning via gradient descent.
 - Normally updates are ΔW (shape $[N \times d]$)
- In LoRA, we freeze W and update instead a low-rank decomposition of W :
 - A of shape $[N \times r]$,
 - B of shape $[r \times d]$, r is very small (like 1 or 2)
 - That is, during finetuning we update A and B instead of W .
 - Replace $W + \Delta W$ with $W + BA$.

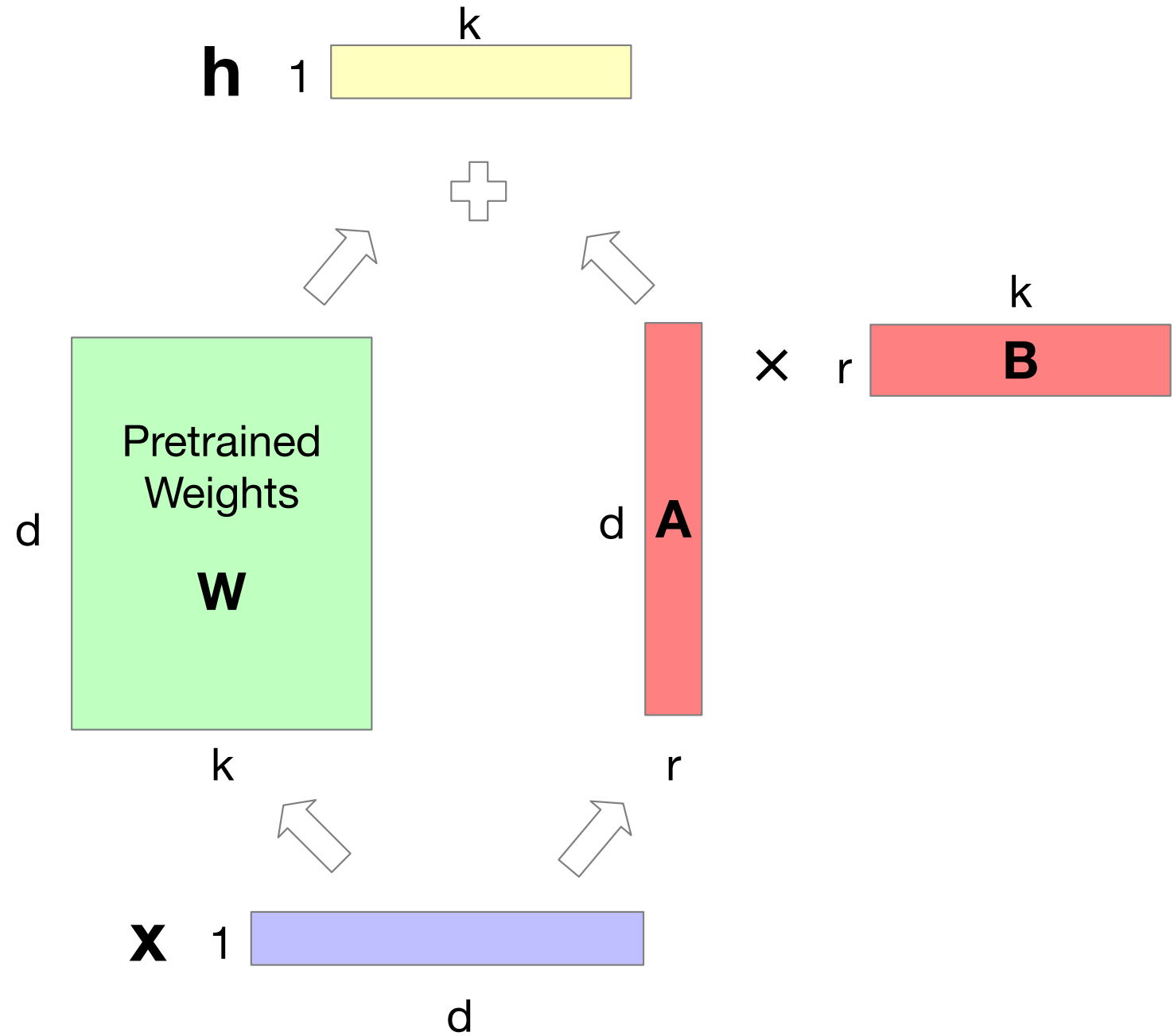
Forward pass: instead of

$$\mathbf{h} = \mathbf{xW}$$

We do

$$\mathbf{h} = \mathbf{xW} + \mathbf{xAB}$$

LoRA



Large
Language
Models

Dealing with Scale

Large
Language
Models

Harms of Large Language
Models

Hallucination

*Chatbots May 'Hallucinate'
More Often Than Many Realize*

What Can You Do When A.I. Lies About You?

People have little protection or recourse when the technology creates and spreads falsehoods about them.

Air Canada loses court case after its chatbot hallucinated fake policies to a customer

The airline argued that the chatbot itself was liable. The court disagreed.

Copyright



Authors Sue OpenAI Claiming Mass Copyright Infringement of Hundreds of Thousands of Novels

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.



Privacy

How Strangers Got My Email Address From ChatGPT's Model

Toxicity and Abuse

The New AI-Powered Bing Is Threatening Users.

Cleaning Up ChatGPT Takes Heavy Toll on Human Workers

Contractors in Kenya say they were traumatized by effort to screen out descriptions of violence and sexual abuse during run-up to OpenAI's hit chatbot

Misinformation

Chatbots are generating false and misleading information about U.S. elections

Large
Language
Models

Harms of Large Language Models